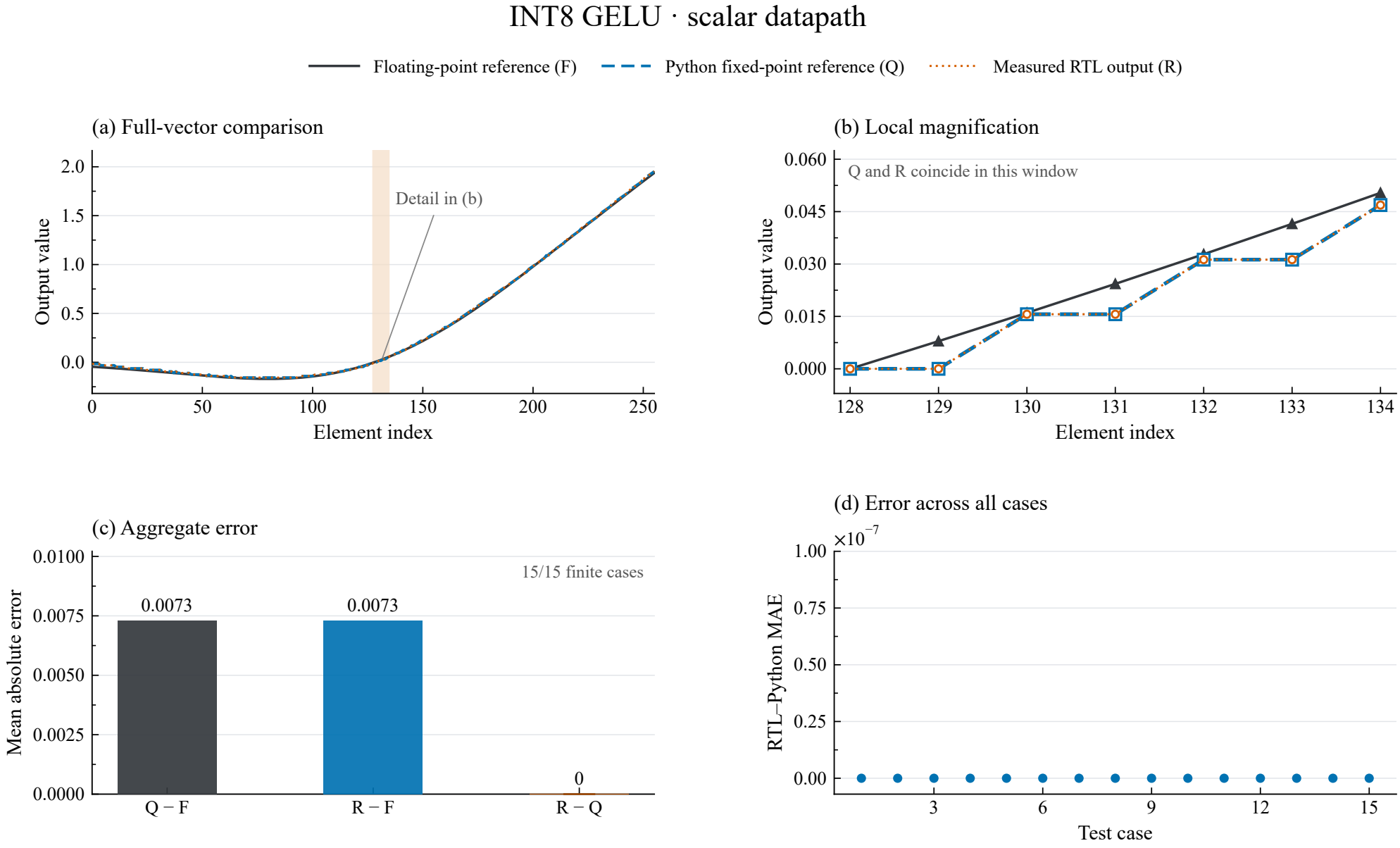
INT8 GELU · scalar datapath
Floating-point reference (F)
Python fixed-point reference (Q)
Measured RTL output (R)
(a) Full-vector comparison
Output value
Element index
Detail in (b)
(b) Local magnification
Q and R coincide in this window
Output value
Element index
(c) Aggregate error
Mean absolute error
15/15 finite cases
0.0073
0.0073
0
Q − F
R − F
R − Q
(d) Error across all cases
×10−7
RTL–Python MAE
Test case
15 exhaustive scale sweeps (Q0–Q14) · 3,840 elements · example Q6 · SIMD backpressure is a separate check.
No vertical offsets. Red markers retain failed checks; all timing values are reported in tables.